



BIMANUAL DINNER-TABLE AI

Two robot arms set the table from one sentence — on Intel edge hardware

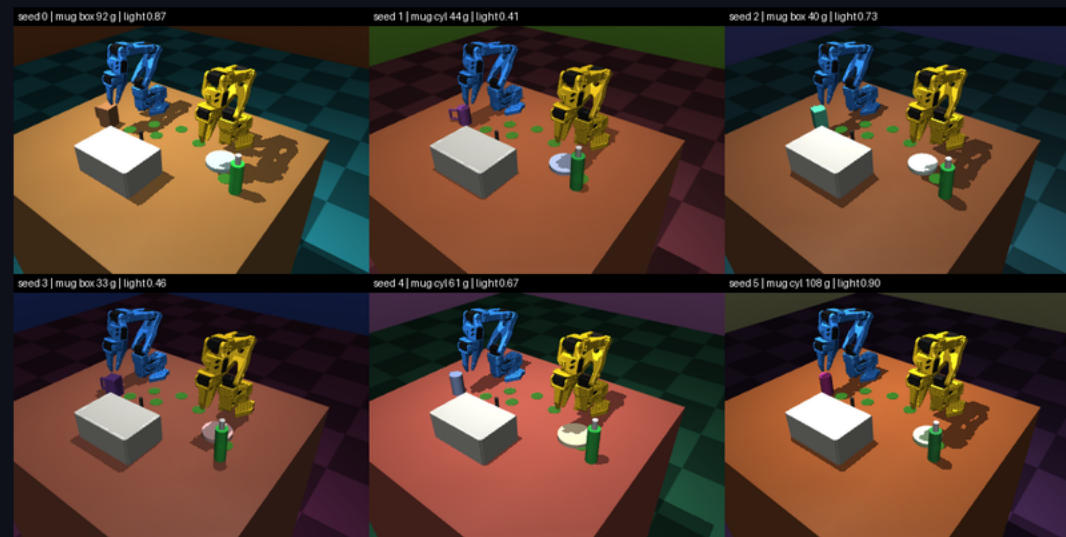
Dual SO-101 · MuJoCo · ACT policy · OpenVINO INT8 1.35 ms on Intel CPU

THE TASK

One command, a full table setting

“Open the top drawer with arm A, pick up the plate ... hand the fork over to arm A ... pour water into the mug with arm A, then put the bottle back and place the mug on the table.”

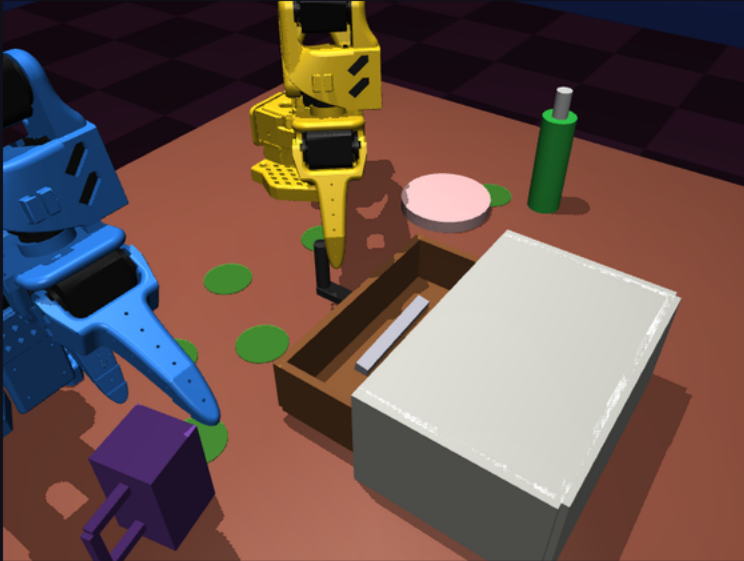
- Two simulated SO-101 arms in MuJoCo must share one workspace
- Multi-step: drawer → cutlery → plate → hand-over → two-arm pour → tidy-up
- Every run randomizes object pose, mass, friction, size, shape and lighting
- Must run perception + policy on Intel edge hardware with OpenVINO



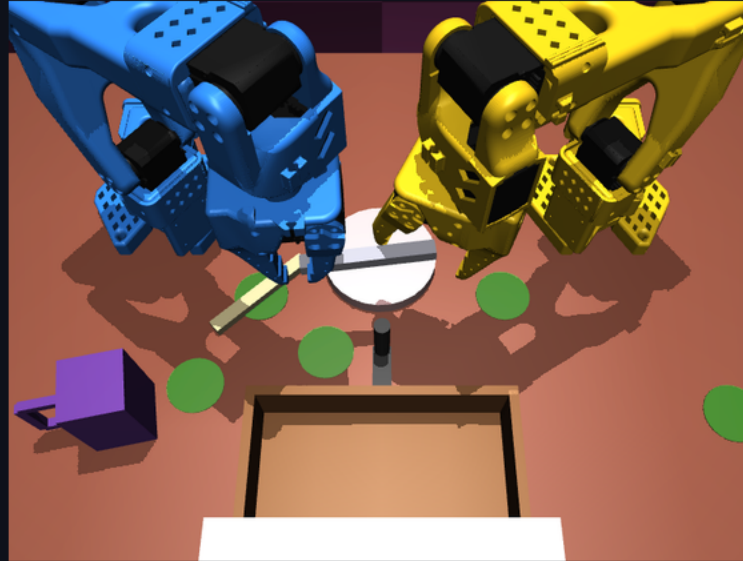
six randomized seeds

WHAT THE ROBOTS DO

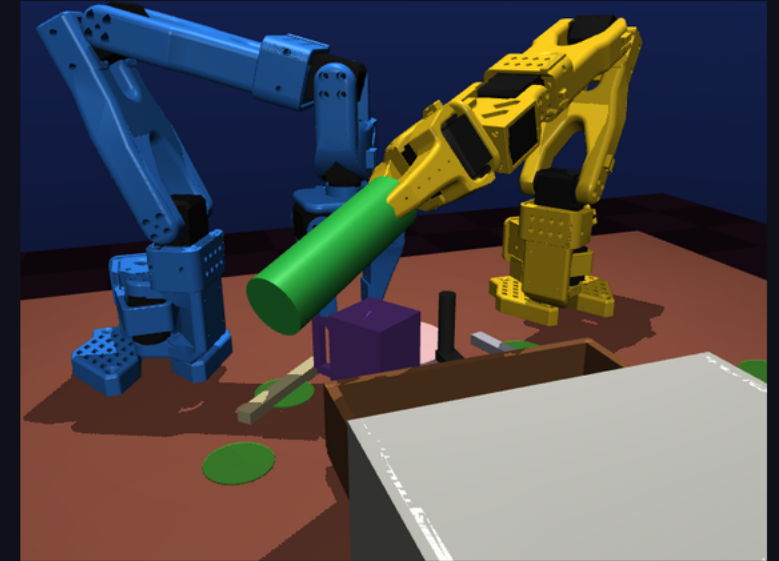
Bimanual coordination



Arm A opens the drawer



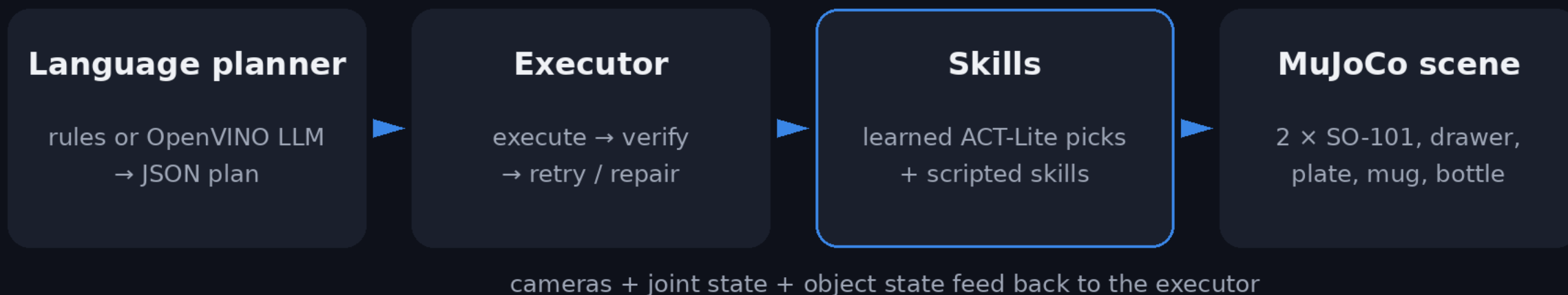
Fork hand-over B → A



B holds the mug, A pours

- Hand-over: receiver grasps the free end first, then the giver releases — no drop
- Pour: holder turns the mug so its handle is out of the pourer's path; staged $40^\circ \rightarrow 90^\circ$ tilt with IK on the spout

Architecture



- Planner tracks what each arm holds, resolves “it”, opens the drawer when cutlery is needed
- Executor remembers achieved goals and inserts repair steps if one is undone (e.g. drawer bumped shut)
- ACT-Lite: 2 cameras + 12-D state + instruction tokens → 20-step action chunk (1.6 M params)
- Scripted expert doubles as the demonstrator: 300 episodes, 117k frames

TASK SUCCESS

Results over 10 randomized seeds

10 / 10

scripted skills, seeds 0-9

7 / 10

held-out seeds 10-19

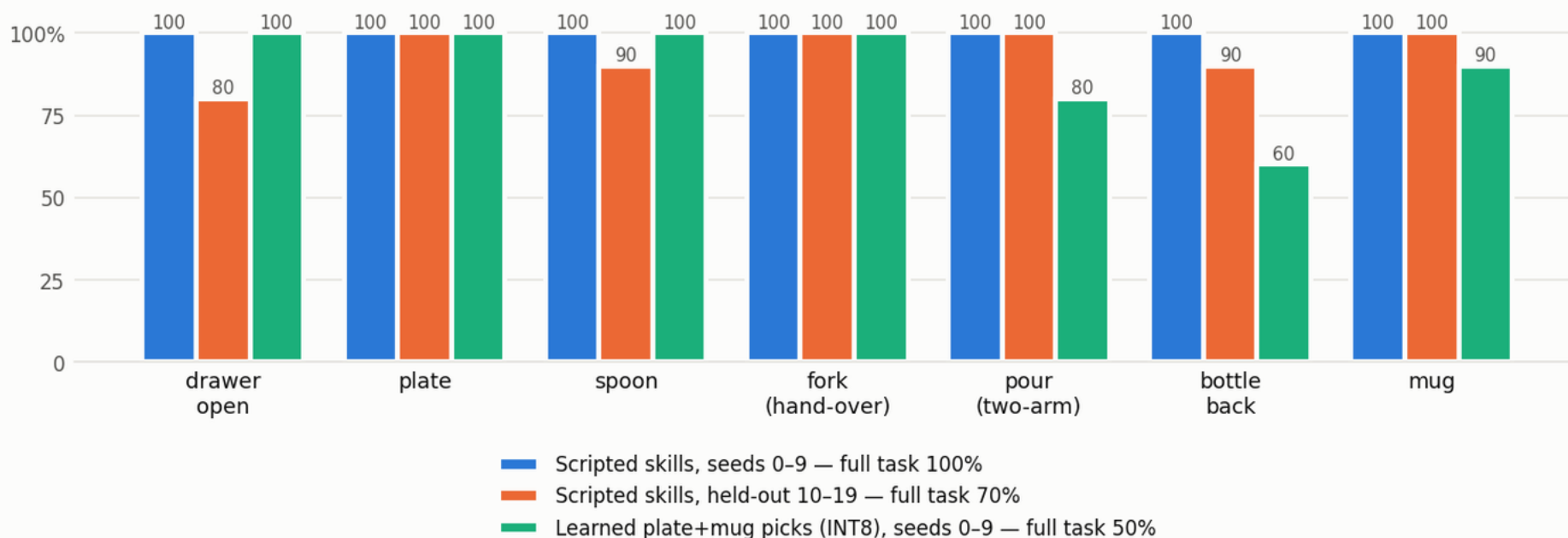
5 / 10

with learned plate + mug
picks

80 %

learned picks solved with no
fallback

Sub-goal success over 10 randomized seeds



OpenVINO on Intel

1.35 ms

INT8 on i9-13900HX CPU

5.2x

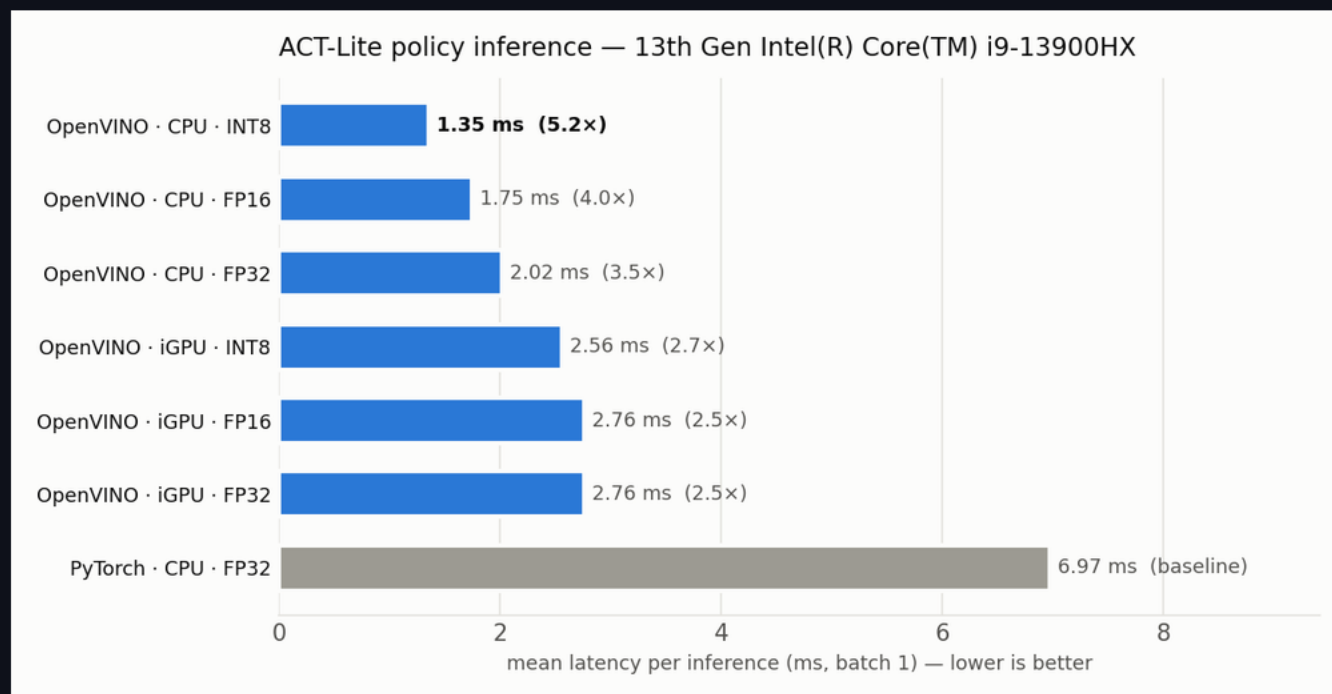
faster than PyTorch

3,421 /s

throughput mode, INT8 CPU

2.56 ms

INT8 on Intel iGPU



- FP32 / FP16 / INT8 IR; INT8 via NNCF post-training quantization
- INT8 output within 0.043 of PyTorch
- In the loop: 6.6 ms per inference vs a 250 ms re-plan budget
- Physics step 1.5 ms + 2-camera render 1.9 ms → faster than real time
- NPU path auto-detected on Core Ultra (test laptop has none)

STATED OPENLY

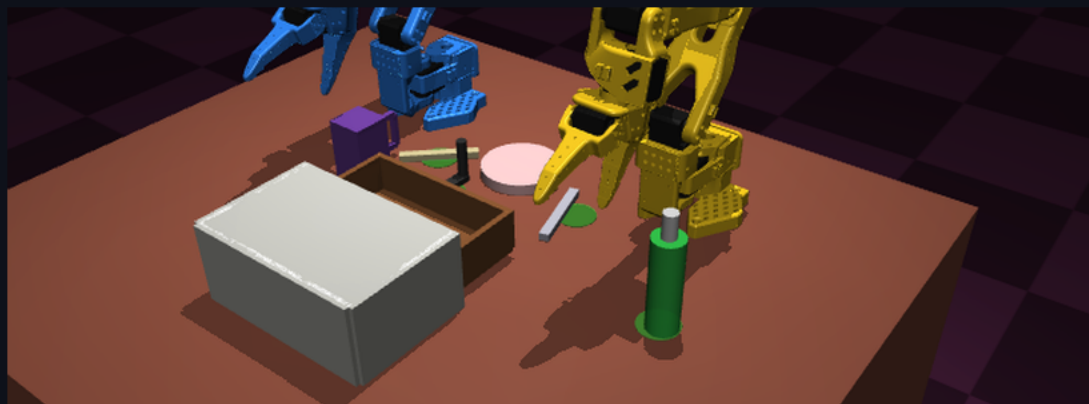
Limitations and next steps

Limitations

- Grasp assist: a weld attaches the intended object once the gripper closes on it
- Pouring is judged geometrically — no fluid simulation
- Learned policy covers picks only; place, hand-over and pour are scripted
- Benchmarked on i9-13900HX (no NPU), not Core Ultra

Next steps

- Train place and hand-over skills; collision-aware loss near the drawer
- Fine-tune SmolVLA / Pi0.5 on the same LeRobot-style data
- Run the benchmark on Core Ultra NPU
- Replace grasp assist with contact-rich grasping



REPRODUCE

Try it

```
$ pip install -r requirements.txt  
$ python scripts/run_episode.py --seed 3 --video videos/seed3.mp4  
$ python scripts/evaluate.py --seeds 0-9 --videos videos  
$ python -m deploy.benchmark --models models/act_pick_int8.xml --sim
```

github.com/HUNT-001/bimanual-table

Code · MuJoCo scene · training · OpenVINO export · benchmark · 10-seed evaluation · video

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